Mohammad Arham

Python Backend Engineer · Data Engineering · Cloud & DevOps

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PROFESSIONAL SUMMARY

Results-driven Python Backend Engineer with 2 years of experience architecting scalable, cloud-native backend systems, high-performance REST APIs, and automated data pipelines. Proficient in FastAPI, Flask, PostgreSQL, AWS (EC2, S3, Lambda, Athena, Redshift), and Apache Airflow. Experienced with LLM integration, AI-assisted development, and GenAI tooling. Expertise in microservices, ETL/ELT pipelines, ERP customization (Odoo), Docker, and CI/CD automation. Delivered 60% database size reduction, cut data latency to 15-minute intervals, and improved pipeline observability by 55%. Passionate about clean code, API security, and scalable system design.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript

Frameworks: FastAPI, Flask, Django REST Framework, SQLAlchemy, Pydantic v2, Odoo ORM, Leaflet.js

Cloud & DevOps: AWS (EC2, S3, Lambda, SNS, SQS, Secrets Manager, Athena, Redshift), Firebase Cloud Messaging (FCM), Docker, Jenkins, Nginx, Certbot, CI/CD

Databases: PostgreSQL, MySQL, Amazon Redshift, Redis, SQLite

Data & Pipelines: Apache Airflow, ETL/ELT, Incremental Sync, Data Ingestion, Task Scheduling, AWS Glue, Google BigQuery

API & Security: REST APIs, JWT, OAuth 2.0, RBAC, Rate Limiting, SSL/TLS, CORS, HTTPS, 2FA

Architecture: Microservices, Async APIs, Event-Driven Design, System Design, DB Query Optimization, Caching (Redis), Distributed Systems, DSA

Tools: Git, GitHub, Linux, Postman, Swagger/OpenAPI, VS Code

WORK EXPERIENCE

Binary Semantics Ltd | Associate Software Developer **Aug 2024 – Present**

Full-time · Gurugram, India · Python, FastAPI, Flask, PostgreSQL, Apache Airflow, AWS, Docker, Odoo

- **Backend & API Development:** Designed and deployed scalable backend services using Flask and FastAPI with PostgreSQL; implemented JWT authentication, RBAC, OAuth 2.0, and API rate limiting to ensure secure, high-performance RESTful endpoints consumed by analytics platforms.
- **Data Engineering & ETL:** Architected a config-driven, incremental ETL pipeline using Python and Apache Airflow to replicate Odoo PostgreSQL data to Amazon Athena and Redshift; reduced data latency from daily batch loads to near real-time 15-minute intervals, enabling BI dashboard freshness for executive reporting.
- **Cloud Infrastructure & Optimization:** Migrated binary file storage from PostgreSQL to AWS S3, shrinking database size by ~60% and significantly improving query response times; deployed all services on EC2 behind Nginx with SSL via Certbot and custom domain routing.
- **ERP Customization (Odoo):** Built and extended Odoo modules using Python ORM; automated background job scheduling with Airflow DAGs and improved ERP workflow reliability, reducing manual intervention and response times for core business processes.
- **Third-Party & Real-Time Integrations:** Integrated ULIP, PAN, and DL verification APIs for KYC and finance compliance; implemented Google Maps API for real-time FASTag vehicle tracking in the logistics module, improving operational visibility.
- **Real-Time GPS Tracking & Notifications:** Built a real-time vehicle tracking pipeline using Google BigQuery for high-frequency GPS ingestion and a Flask/Leaflet.js web map for live, customer-facing shipment tracking, with push-notification delivery (AWS SNS/FCM) to iOS and Android driver apps; optimized the map for production scale, cutting page load time from 1.5s to under 15ms and map-rendering overhead by ~240x on long routes.
- **DevOps & CI/CD:** Containerized all backend services with Docker; built Jenkins CI/CD pipelines for automated testing and zero-downtime deployments; managed secrets via AWS Secrets Manager; automated database dump workflows via Airflow with AWS SNS failure alerts.

- **AI & Chatbot Integration:** Integrated a Rasa NLU-powered chatbot with core backend APIs via REST webhooks; mapped user intents to business logic actions, reducing manual support volume and accelerating query resolution time.

PROJECTS

Real-Time GPS Live Tracking System

Python · Flask · Google BigQuery · AWS SNS/FCM · Leaflet.js

- Built a real-time vehicle tracking pipeline using Google BigQuery for high-frequency GPS ingestion and a Flask-based web map (Leaflet.js) for live, customer-facing shipment tracking, with a Python backend handling push-notification delivery (AWS SNS/FCM) to iOS and Android driver apps.
- Optimized the tracking map's backend and frontend for production scale — cut page load time from 1.5s to under 15ms and reduced map-rendering overhead by ~240x for long routes, validated with automated end-to-end tests against live production data.

Inlog Analytics Dashboard — Enterprise BI Platform

Flask · PostgreSQL · Raw SQL · CanvasJS · JavaScript · REST APIs · Docker · AWS EC2 · RBAC

- Engineered a multi-module Flask backend serving RESTful APIs to a dynamic frontend; implemented a session-based filtering engine for real-time data segmentation across 5 business units (Sales, Marketing, HR, Logistics, Helpdesk) with RBAC-enforced role-specific access.
- Authored and optimised complex PostgreSQL queries with CTEs, window functions, and aggregations to transform raw Odoo ERP data into drilldown KPIs and trend benchmarks; deployed on AWS EC2 with Docker behind Nginx.

FastAPI Microservice — Real-Time Analytics API

FastAPI · Async Python · Pydantic v2 · PostgreSQL · AWS EC2 · Nginx · Docker

- Engineered a high-throughput async FastAPI microservice powering real-time analytics dashboards; leveraged Pydantic v2 schema validation, PostgreSQL connection pooling, and optimized indexed queries to achieve sub-50ms p95 response times.
- Deployed behind Nginx on AWS EC2 with HTTPS and health-check endpoints; reduced average database query time by ~40% through selective indexing and query plan analysis.

Config-Driven ETL Pipeline — Apache Airflow & AWS Athena

Python · Apache Airflow · AWS Athena · Redshift · PostgreSQL · AWS SNS

- Designed a modular, YAML-config-driven ETL framework with Airflow DAGs supporting incremental sync, full backfill, and automatic schema evolution for upstream Odoo data into AWS Athena and Redshift.
- Added SLA breach alerting via AWS SNS and Airflow callbacks; improved data-team incident response time by ~60% — pipeline now serves as the primary data source for 3 internal BI dashboards.

EDUCATION

Jamia Hamdard University — B.Tech in Computer Science **2020 – 2024 | CGPA: 8.19 / 10.0**

New Delhi, India · Relevant: Data Structures & Algorithms, Database Management, Operating Systems, Software Engineering, Computer Networks

CORE COMPETENCIES & KEYWORDS

Python • Backend Development • REST API • FastAPI • Flask • Microservices • Cloud Computing • AWS • Docker • CI/CD • Jenkins • DevOps • Data Engineering • ETL Pipeline • Apache Airflow • Google BigQuery • PostgreSQL • MySQL • Amazon Redshift • Amazon Athena • S3 • EC2 • Lambda • JWT Authentication • LLM Integration • OAuth 2.0 • RBAC • API Security • System Design • Database Optimization • Async Python • SQLAlchemy • Pydantic v2 • Odoo ERP • Git • Linux • Nginx • AI Integration • NLP • Distributed Systems • Real-Time Analytics • GPS Tracking • Leaflet.js • Push Notifications • Firebase Cloud Messaging • Agile • Scrum